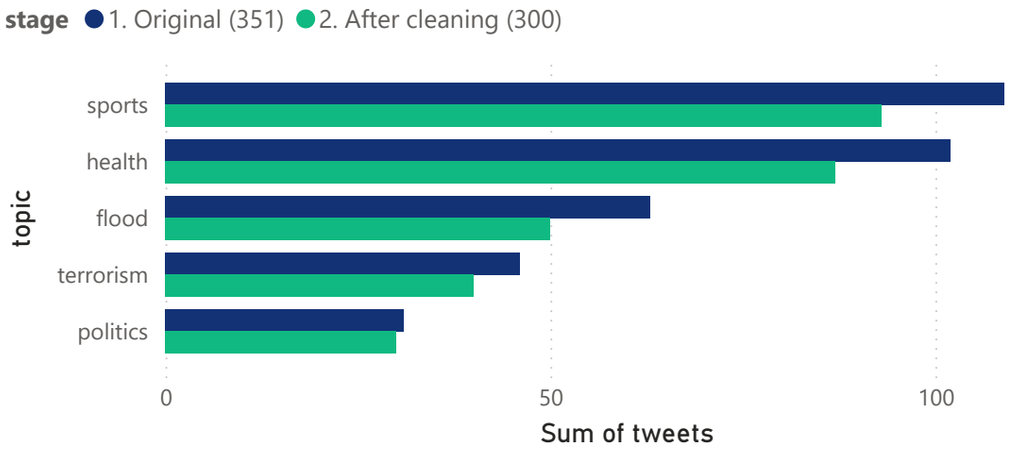
# Twitter Topic Classification & Pattern Mining

Overview | Architecture Diagram | Pattern mining | Algorithms | Performance | Architecture

| 351 | 300 | 5 | 86.67% |
|---|---|---|---|
| Original tweets | Tweets after cleaning | Topic classes | Best test accuracy |

## Class distribution: original vs cleaned




## Signature words per topic

| topic | Top words |
|---|---|
| flood | flood, kerala, home, twitter |
| health | health, care, existing, think |
| politics | house, politics, white, deputy |
| sports | sports, nadal, kaepernick, usopen |
| terrorism | terrorism, anniversary, america, government |

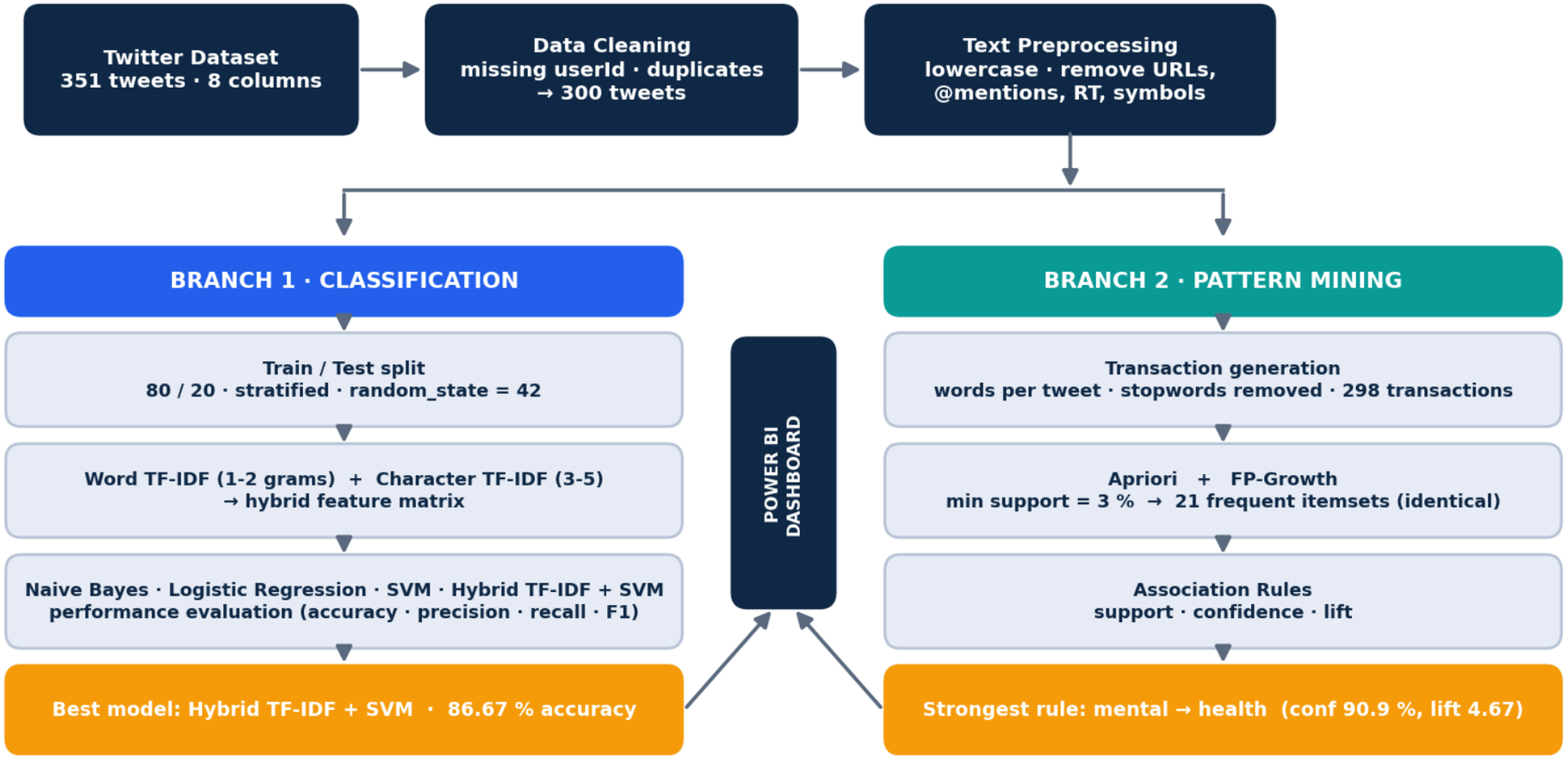
## Input: tweet text → Output: topic label

| Input tweet | predicted | actual | result |
|---|---|---|---|
| @funder @TheDemocrats @TheDemCoalition @IndivisibleTeam @Ind... | flood | flood | Correct |
| @realDonaldTrump America believes Bob Woodward above your fl... | flood | flood | Correct |
| Flood ins rise faced by many incl Sandy victims – Hard @FEMA... | flood | flood | Correct |
| Flood of Sand Points to Shakeout for Shale Suppliers https:/... | flood | flood | Correct |
| RT @ahmedpatel: I thank the @MahilaCongress leadership and w... | flood | flood | Correct |
| RT @amritabhinder: These flood victims don't make news Like ... | flood | flood | Correct |
| RT @GordonTropical: DON'T GET TOO CUMFY BITCH CAUSE I JUST M... | flood | flood | Correct |
| RT @Mye_Pixie: Be kinder to yourself. And then let your kind... | flood | flood | Correct |
| RT @SfcThalassery: Our contribution to Kerala Flood relief c... | flood | flood | Correct |
| RT @Franktmcveety: Joshua Boyle Accused of Abuse and Ideolog... | flood | politics | Wrong |
| $TWIST can't protect you from an #STD but w/a Web Wallet w/e... | health | health | Correct |
| @delwilliams @shomaristone Is the increase in therapy direct... | health | health | Correct |
| @Health_Reforms Need to be self funded and more important pa... | health | health | Correct |
| @SnowBiAuthor @KattMcC @Justice42557649 @cdahlgrd @MNarvesta... | health | health | Correct |
| A review of the role of #tau protein in health and disease: ... | health | health | Correct |

**60 test tweets · 52 correct · 8 wrong**

# Twitter Topic Classification & Pattern Mining

## System Architecture: Twitter Topic Classification & Pattern Mining



# Twitter Topic Classification & Pattern Mining

Overview | Architecture Diagram | Pattern mining | Algorithms | Performance | Architecture

Minimum support

3.00%

**21**
Itemsets found

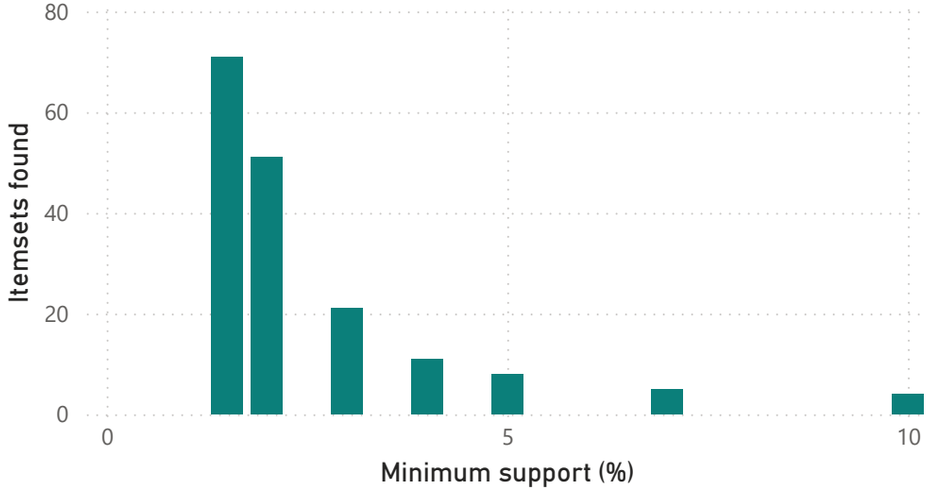
**2**
Rules found

**4.67**
Strongest lift

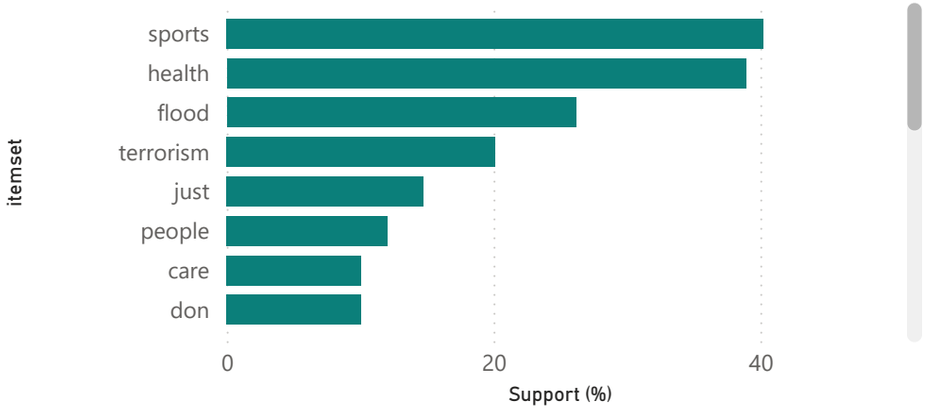
## Association rules (confidence ≥ 50%)

| rule | Support | Confidence | Lift |
|---|---|---|---|
| care → health | 4.03 | 80.00 | 4.11 |
| mental → health | 3.36 | 90.91 | 4.67 |

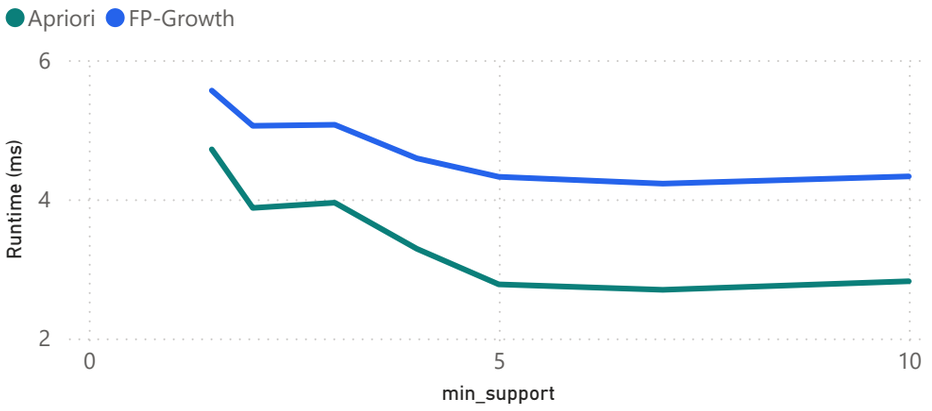
## Itemsets found vs minimum support




## Most frequent items (support %)



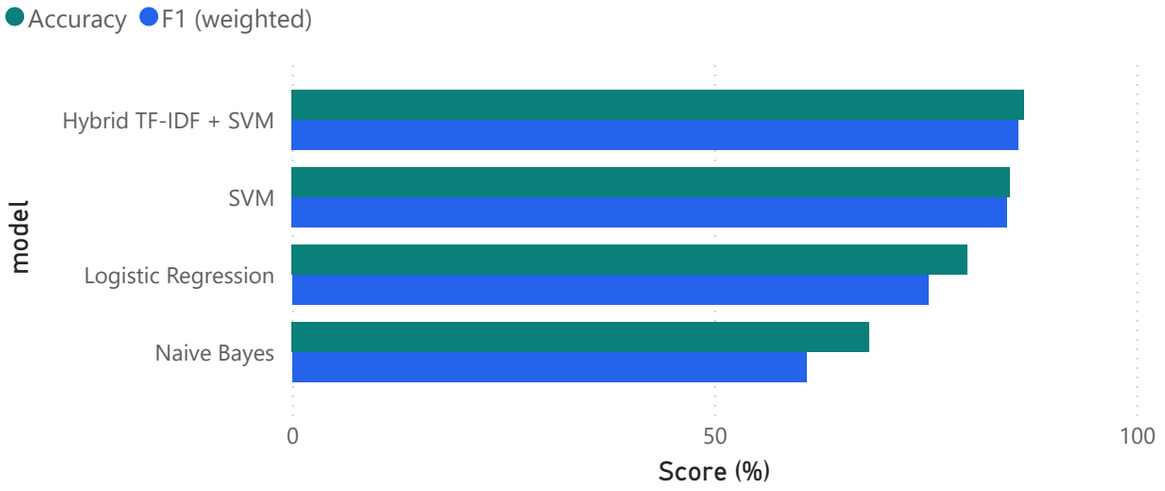

## Apriori vs FP-Growth runtime (ms)



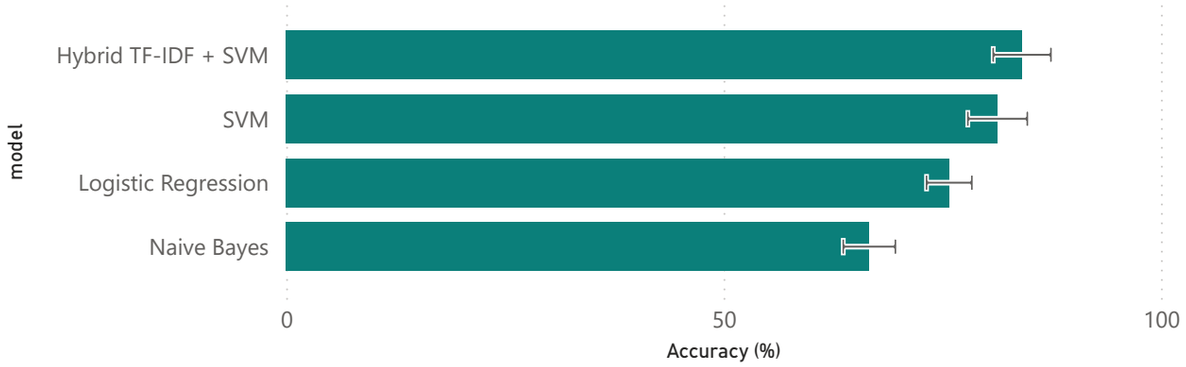
# Twitter Topic Classification & Pattern Mining



## Test set (60 tweets): accuracy and F1 (%)

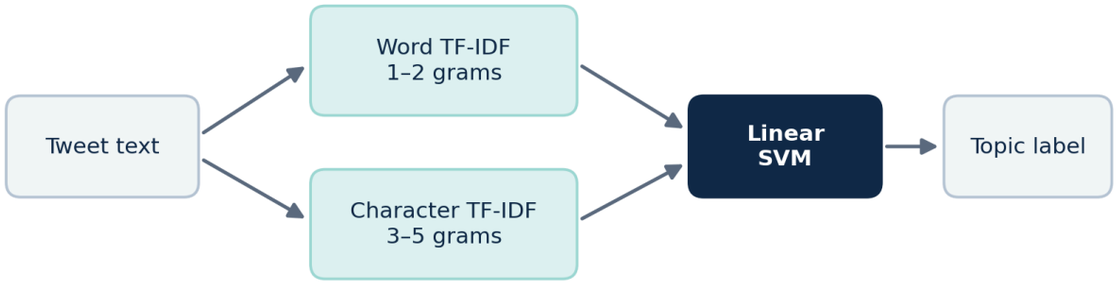



## Cross-validation accuracy (5-fold × 5 repeats, 300 tweets)




**Innovation: adding character TF-IDF to the SVM. The hybrid beat the plain SVM in 19 of 25 validation runs (4 lost, 2 tied). Wilcoxon p = 0.0003**

## Hybrid TF-IDF + Linear SVM




Words capture meaning; character patterns catch spelling variants and hashtags.

| | | |
|---|---|---|
| **85.00** SVM (test) | **86.67** Hybrid (test) | **1.67** Gain (test) |
| **81.33** SVM (CV) | **84.13** Hybrid (CV) | **2.80** Gain (CV) |

# Twitter Topic Classification & Pattern Mining

Overview | Architecture Diagram | Pattern mining | Algorithms | **Performance** | Architecture

**Select model**

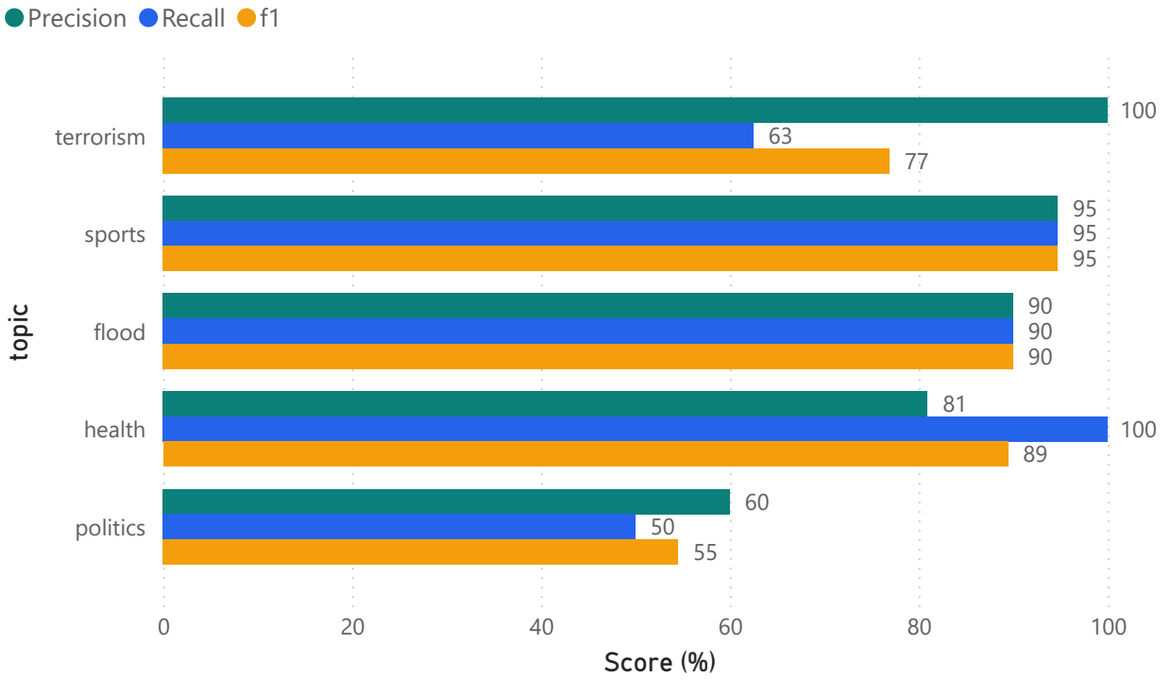
Hybrid TF-IDF + SVM

| Accuracy (%) | Precision, weighted (%) | Recall, weighted (%) | F1 score, weighted (%) |
|---|---|---|---|
| 86.67 | 87.27 | 86.67 | 86.06 |

## Confusion matrix (60 test tweets)

| actual | flood | health | politics | sports | terrorism |
|---|---|---|---|---|---|
| flood | 9 | 0 | 0 | 1 | 0 |
| health | 0 | 17 | 0 | 0 | 0 |
| politics | 1 | 2 | 3 | 0 | 0 |
| sports | 0 | 0 | 1 | 18 | 0 |
| terrorism | 0 | 2 | 1 | 0 | 5 |

## Per-class precision, recall and F1 (%)



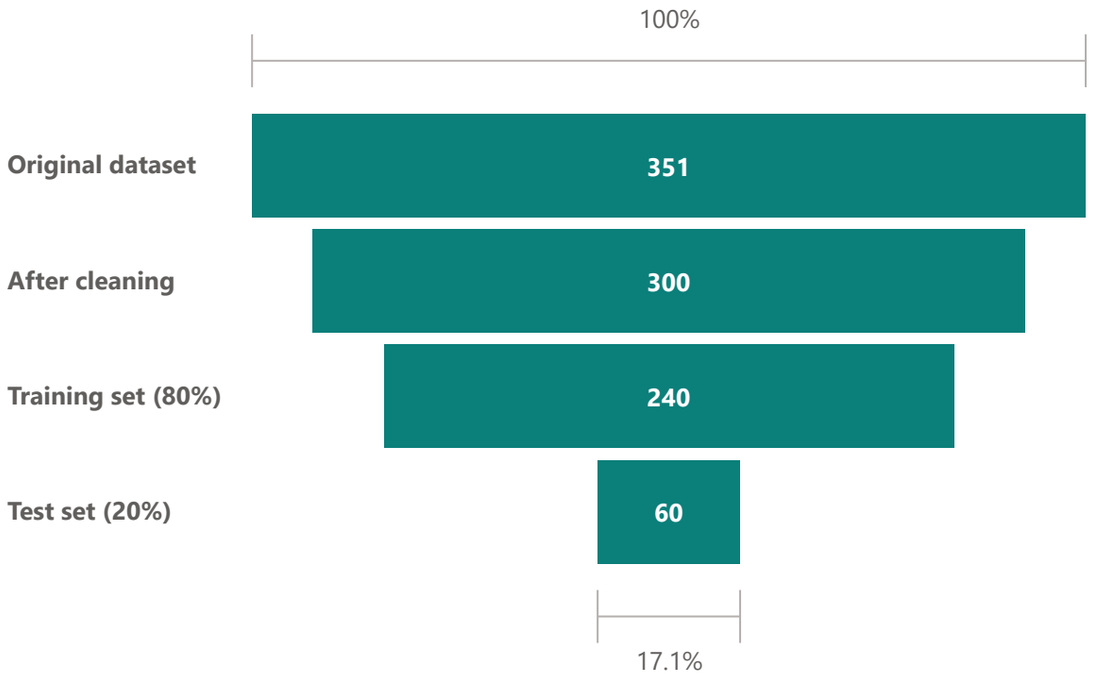

Rows = actual topic · columns = predicted topic · 52 of 60 on the diagonal (Hybrid)

Politics is the weakest class: only 6 test tweets, mostly confused with health

# Twitter Topic Classification & Pattern Mining



## Rows at each pipeline stage




| Classification | Pattern mining | Shared |
|---|---|---|

## Train / test split by topic

